What the experiments show, and what they do not

A walk through the REMORA experiments for readers who do not know the code base. Each experiment is described the same way: the question it asked, how it was tested, what was measured and what the result means.

The idea being tested

A tool-using AI agent can act in the world: read customer records, send an email, pay an invoice. REMORA lets the agent *propose* an action but never lets it grant itself the authority to carry it out. Authority comes from signed rules the deploying organisation owns, and a fixed, rule-based gate checks each call before it runs.

`agent proposes` → `gate decides` → `approval for this exact call` → `executed` → `effect checked`

Every experiment measures two quantities that pull against each other.

Harm let through

A harmful or wrong call that was executed. This should be as close to zero as possible.

Useful work let through

A legitimate call that was allowed to run. When this is low, the system is safe only because it refuses almost everything.

A gate that refuses everything scores zero on both. Several experiments therefore compare REMORA with a refuse-all control.

EXPERIMENTS

1. Synthetic tool-call benchmark
2. AgentHarm
3. BFCL C-ext3: valid call, wrong task
4. BFCL C-ext4: a revised rule bundle
5. WS1: bounded lookups when the gate is unsure
6. Attacks on the execution chain
7. AgentDojo: first run on unseen tools

useful work let through · harmful or wrong call let through · target or reference

700 rows, effectively 70 independent cases · CLAIM-001, CLAIM-020

1. Synthetic tool-call benchmark

QUESTION

Does the gate let harmful tool calls through?

HOW IT WAS TESTED

Constructed tool calls, harmful and benign. Many rows are variants of one template, so the number of independent cases is 70.

REMORA, harmful templates accepted	0 of 56
Keyword baselines, unsafe accepted	1.4%

> The gate works, but it is not shown to beat a simple keyword rule. The difference is not significant, and an ablation attributes the result to no single component.

Zero of 56 still allows a true failure rate of up to 6.4% (Wilson 95% upper bound).

208 harmful scenarios with benign counterparts · CLAIM-002

2. AgentHarm

QUESTION

Can the gate tell a harmful task from a near-identical benign one?

HOW IT WAS TESTED

A public benchmark of harmful agent tasks built by external researchers. Each harmful task has a benign counterpart that looks alike.

Harmful tasks authorised	0%
Benign counterparts authorised	0%

> The run authorised none of the harmful tasks and none of the benign ones. It does not discriminate; it refuses.

The paper also reports a separate tri-mode AgentHarm run on 88 tasks through a different decision path. The two are different experiments, not one benchmark at two sample sizes.

sealed, pre-registered · BFCL v4 · CLAIM-019

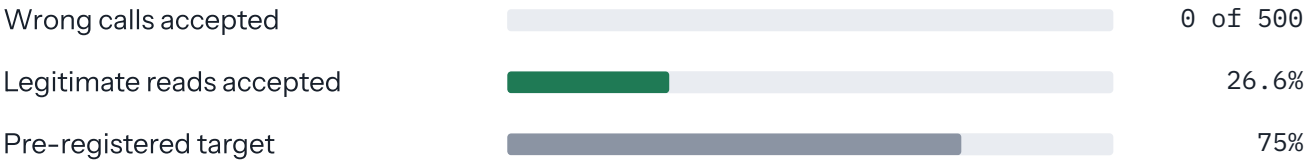
3. BFCL C-ext3: valid call, wrong task

QUESTION

A call can be well-formed and still wrong for the task. Does the gate notice?

HOW IT WAS TESTED

A correct call from one task was substituted into another. The gate received declared tool contracts and a task intent derived from the task text. The plan and its targets were fixed before the data was opened, and the track was run once.



> Very safe and too strict. The joint hypothesis of high safety and high utility was not supported: four of seven pre-registered targets were missed.

One missed target is a safety target: 2 of 199 constructed wrong-tool calls were accepted (1.005% against a bar of at most 1%). The wrong calls were substituted by the researchers, not produced by an agent.

sealed · 1,562 episodes · CLAIM-021

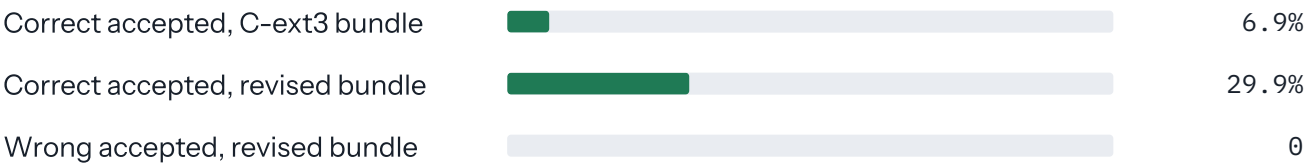
4. BFCL C-ext4: a revised rule bundle

QUESTION

Can better rules let more correct calls through without letting wrong ones through?

HOW IT WAS TESTED

The last unspent BFCL v4 reserve. The C-ext3 bundle and the revised bundle ran on the same episodes, next to a refuse-all control.



> A measurable gain with no wrong call accepted. Seven in ten correct calls are still refused.

The gain sits on tools seen during development. Transfer to new tool families is untested.

sealed · 825 episodes, 11 synthetic ISP customers · NEGATIVE_RESULTS §95, §96

5. WSl: bounded lookups when the gate is unsure

QUESTION

When the gate is unsure, can the agent fetch a small, authorised fact, such as the customer's work order, and get the call approved without the model deciding?

HOW IT WAS TESTED

After the lookup, the same gate decides again from scratch. Five arms were compared: no lookups, fixed-order lookups, cost-ordered lookups, lookups prioritised by a learned monitor, and no gate at all. Attacks ran in the same study.



Bars show benign reads completed safely. Without a gate all 33 completed, but five attack goals executed, with seven leak episodes and seven out-of-work-order episodes. No gated arm showed any of these in 121 attack episodes each.

> Verification recovers much of the utility without letting the model define its own authority. The learned monitor looked better, but it ran at a looser veto threshold (0.4 against 0.3). That explains the difference; better prioritisation is not shown.

The earlier Study B showed the same pattern: fixed-order lookups 28 of 45, no lookups 0 of 45, the monitor 15 of 45. Neither holdout attributes a gain to the learned monitor. The customers are synthetic and the author built every part.

post hoc, on the author's own tool catalogue · paper, Negative Results

6. Attacks on the execution chain

QUESTION

Does the chain from approval to effect hold under attack, and what still gets through?

HOW IT WAS TESTED

Defined attacks on the chain, run against REMORA and against a gateway that only checks policy. A separate group tried to move a secret out through a sequence of calls that are each allowed.



> Checking each call on its own does not stop a leak assembled from individually allowed calls. This is the most important boundary the experiments found.

With an opt-in flow check, REMORA ran none of the 10 exfiltrations it was tested on, but only 2 of 4 benign controls. The context labels were assigned by the harness, not by a real deployment.

development dry run, not confirmatory · 29 tasks, 91 calls · NEGATIVE_RESULTS §92, §93

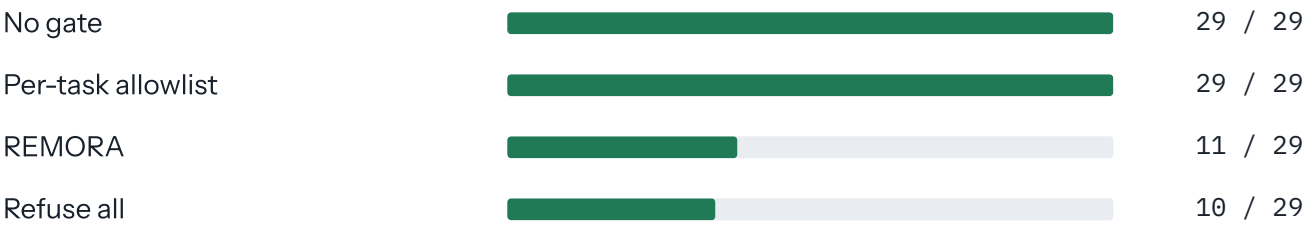
7. AgentDojo: first run on unseen tools

QUESTION

Do the rules work outside the tool catalogue they were developed on?

HOW IT WAS TESTED

The ground-truth calls of 29 AgentDojo development tasks were replayed through each arm by a scripted agent, with no model and no attacks. Every call was legitimate, so every block cost useful work.



Bars show tasks completed. REMORA blocked 90 of 91 legitimate calls. The allowlist is built from each task's ground truth, which a deployment would not have.

> The semantic part of the gate did not transfer to unseen tools. Here REMORA completed one task more than refusing everything, and a bounded lookup could not have recovered any blocked call (§93).

The overall picture

SHOWN	NOT SHOWN
Binding approval to one exact call stops the defined attacks on the execution chain.	That REMORA is safer than existing defences on the same tasks.
The gate can refuse calls that are valid but wrong for the task, on tools it was developed on.	That this transfers to new tools. The AgentDojo dry run points the other way.
Bounded, authorised lookups recover utility without the model deciding.	Any gain from a learned monitor.
Negative results are published and tied to stored artifacts.	Independent replication, which is still open (REM-021).

The next experiment

Can labels that follow the data flow replace lexical matching of task intent, so that the gate keeps its utility on unseen tools while injected attacks still fail?

1. **Development:** let reads run under information-flow labels, and show on the 30 AgentDojo development tasks that REMORA clearly beats refuse-all.
2. **Pre-registration:** fix the plan, agent model, monitor prompt and success criteria on OSF before the test split is opened.
3. **Sealed run:** a real language-model agent, AgentDojo's built-in injection attack, and seven defences side by side: no gate, refuse-all, per-task allowlist, an LLM monitor, REMORA and two compositions. The result is reported as a safety–utility frontier.

Either outcome is publishable. A pass is preserved utility at equal safety on an external benchmark where other defences have published numbers. A fail confines the contribution to exact-call binding.

Sources: `paper/remora_paper.tex`, `docs/assurance/claim_register_v1.yaml` and `NEGATIVE_RESULTS.md` in REMORA-research. Numbers are copied from them unchanged.

All results are the author's own, and most use synthetic or constructed data. They are not a production-security guarantee.